

1.6 Need Analysis for Interface Design

نیڈ اینالیسس موجودہ صورتحال اور مطلوبہ نتیجے کے فرق کو یو آئی بنانے سے پہلے تلاش کرتا ہے۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 09 of 15

1.6 Need Analysis for Interface Design

LEARN IT

Need analysis is the process of identifying the gap between the **existing situation** and the **desired outcome**. In HCI it is done **before** designing a UI, so that you understand users' goals, tasks, abilities and expectations.

If you skip need analysis, you guess. Guessing is how a school portal ends up with 40 menus and no "Download date sheet" button on the home screen.

Steps in need analysis:



1. **Identify the users:** who exactly will use the system? (elderly people, teenagers, ...)
2. **Information collection:** gather data using interviews, questionnaires or observations.
3. **Analyse user needs:** identify the core problems, goals and requirements.
4. **Prioritise needs:** decide which features are the most critical to build first.
5. **Develop solutions:** design a system that meets those verified needs.

WORKED EXAMPLE

Case study from the source: mobile banking app. Before building the app, the team interviews customers and discovers:

- Users want secure login via Fingerprint / FaceID.
- Elderly users need large buttons and simple navigation.
- Users want support for both English and local languages.

Result: the UI is designed around these proven needs, which makes success more likely.

Apply the five steps to a college portal in Sindh.

1. Users: Grade XI–XII students, parents, clerks, teachers. Not "everyone in Pakistan".
2. Collect: interview 20 students; questionnaire to parents; watch a queue at the office on fee day.
3. Analyse: students cannot find date sheets; parents cannot read English-only fee notices; clerks re-type the same form.
4. Prioritise: (1) date sheet on home screen, (2) Urdu/Sindhi fee notice, (3) online leave form. Fancy animations wait.
5. Solution: a simple portal with those three features, large buttons, bilingual labels.

REMEMBER IT

Need analysis = gap finding. Write: Existing = what hurts now. Desired = what "done" looks like. The UI must close that gap, not decorate it.

COMMON MISTAKE

Mistake: Starting in Figma on day one, then "asking users later".

Fix: Need analysis comes **before** UI design. Features you did not hear from users are guesses. Guessed features waste time and fail in exams when the question is "why do we analyse needs?"

اردو میں سمجھیں

نیڈ اینالیسس موجودہ صورتحال اور مطلوبہ نتیجے کے فرق کو یو آئی بنانے سے پہلے تلاش کرتا ہے۔ پانچ قدم: صارفین کی پہچان، انٹرویو یا مشاہدے سے معلومات، ضروریات کا تجزیہ، اہم خصوصیات کو ترجیح، پھر تصدیق شدہ ضرورتوں پر حل۔ موبائل بینکنگ کی مثال: فنگر پرنٹ لاگ ان، بڑے بٹن، اور انگریزی کے ساتھ مقامی زبان۔

CHECK YOURSELF

1. When in the design process is need analysis done?
2. List the five steps in order.
3. Name three needs found in the mobile-banking case study.

Answers: 1. Before designing the UI. 2. Identify users; collect information; analyse needs; prioritise; develop solutions. 3. Fingerprint/FaceID login; large buttons and simple navigation for elderly users; English plus local languages.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Need analysis finds:

- a. The gap between now and the desired UI outcome
- b. The CPU benchmark
- c. The NAND delay
- d. The sprint velocity only

MCQ 2. You study:

- a. Users, tasks, pain points, constraints
- b. Only colours you like
- c. Only competitors' logos
- d. Only Big O

MCQ 3. A persona is:

- a. A realistic portrait of a user type
- b. A primary key
- c. A pie chart
- d. A gate

MCQ 4. Skipping need analysis causes:

- a. Pretty UIs that solve the wrong problem
- b. Correct Big O automatically
- c. Free RAM
- d. Shorter OSI

MCQ 5. Constraints include:

- a. Time, devices, literacy, connectivity
- b. Only Figma seats
- c. Only NAND trainers
- d. Only exam memes

MCQ 6. Output of need analysis is:

- a. Requirements for the interface
- b. A compiled kernel
- c. A K-map
- d. A histogram of NAND

Extra short questions

1. Define need analysis in HCI.
2. Three techniques.
3. Give a college example.
4. What is a constraint here?

Extra long question

1. Do a mini need analysis for a hostel complaint app: users, top 3 tasks, environment, constraints, 5 requirements.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Identify the gap between the current situation and what users need the interface to do.
2. Interview, observe, survey, task analysis, look at old system logs.
3. Students miss fee deadlines → need SMS reminders + a 3-step mobile payment UI.
4. Many students on low-end Androids and 2G — no huge animations.

Long

1. Users: residents, warden. Tasks: report, track, urgent leak. Env: phone on the stairs.
Constraints: Urdu, photos, weak wifi. Req: offline draft, photo, status, privacy, large buttons.

Next lecture: 1.7 Human Computer Interaction Problems